

PARENTHETIC REMARKS

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Abstract: Simple kinds of operads, monads, monoidal categories, free examples of such, and appropriate string diagrams leading to forests, are discussed and related.

Introduction

The Tamari Festschrift project has alerted several of us to the papers of Dov Tamari and to our common interests . . . except that his work came earlier.

My contribution here is a slightly revised version of an incomplete preprint from 1997. While there is little new in what I have to say, except perhaps about the way the topics interact, it is very much in the spirit of Tamari's work. In particular, I move from the example of parenthesising to more general algebraic operations. I invoke a criterion for when an expression involving operations and constants is well formed. By moving from words to cycles, Tamari [Ta]¹ provides added insight to this criterion, aiding enumeration.

My purpose is to provide a gentle introduction to some contemporary mathematical concepts by starting with the parenthesizing example. I emphasise categorical aspects, however, a first course in category theory should be sufficient prerequisite; for example, the concepts required could be found in [ML]. We look at simple kinds of operads, monads, monoidal categories, free ones, and appropriate string diagrams which lead to forests.

We will not look at the related topic of higher categories. However I must say, soon after the appearance of [St1], Jim Stasheff pointed out the similarity between the “oriental” diagrams therein with associahedra. Also, before any precise definitions had appeared, Todd Trimble [Tr] proposed the relevance of operads to defining weak higher categories. Michael Batanin in [B2] and [B3] developed a theory of higher operads which achieved one definition of higher category and I recommend [B1] and [B3] for further reading on these matters.

1. Parengbras

The bracketing² of a string $abcde$ of entities a, b, c, d, e should produce a new entity $(abcde)$. This example shows bracketing as an operation of arity 5. We do not allow $()$ or (a) . This leads us to the following kind of universal algebra. A *parengebra* is a set A together with, for all $n \geq 2$, exactly

¹ How did I miss this paper, published in an Australian journal, dedicated to Hanna Neumann, and communicated by Mike Newman? At the time I was not ready to absorb what it had to say.

² We use the word “bracket”, interchangeably with “parenthesis”, for the symbols $)$ and $($. The former word conjugates better. For us, the symbols $]$ and $[$ are “square brackets”.

one n -ary operation

$$\beta_n : A^n \longrightarrow A \quad (1.1)$$

subject to no axioms. Let $\mathcal{K}$ denote the category of parengbras and their morphisms. The general associativity law implies that every semigroup is canonically a parengbra. In particular, we have the parengbra $\mathbb{N}$ of natural numbers under addition where

$$\beta_n(m_1, \dots, m_n) = m_1 + \dots + m_n. \quad (1.2)$$

Write $F(X)$ for the free parengbra on the set X . The elements of $F(X)$ are built up iteratively as follows:

- i. each $x \in X$ is in $F(X)$;
- ii. if $a_1, \dots, a_n \in F(X)$ for $n \geq 2$ then $\beta_n(a_1, \dots, a_n) \in F(X)$.

Writing $(a_1 \dots a_n)$ for $\beta_n(a_1, \dots, a_n)$, we can imagine elements of $F(X)$ to be words in X (that is, elements of the free monoid X^* on X) with brackets meaningfully inserted. This provides the left adjoint functor

$$F : \mathbf{Set} \longrightarrow \mathcal{K} \quad (1.3)$$

to the forgetful functor $U : \mathcal{K} \longrightarrow \mathbf{Set}$ into the category $\mathbf{Set}$ of small sets. Let

$$K = U \circ F : \mathbf{Set} \longrightarrow \mathbf{Set} \quad (1.4)$$

be the monad on $\mathbf{Set}$ generated by the adjunction $F \dashv U$. As with any forgetful functor from a category of universal algebras, the functor $U : \mathcal{K} \longrightarrow \mathbf{Set}$ is monadic; we have an isomorphism of categories

$$\mathcal{K} \cong \mathbf{Set}^K \quad (1.5)$$

where $\mathcal{K} \cong \mathbf{Set}^K$ is the category of K -algebras for the monad K in the sense of Eilenberg-Moore [EM] (although they used the term “triple” for “monad”).

Consider the functor $R : \mathbf{Set} \longrightarrow \mathbf{Set}$ defined by the power series

$$R(X) = \sum_{n \geq 2} X^n. \quad (1.6)$$

As with any endofunctor, an R -algebra is an object A of the category ($\mathbf{Set}$ in this case) together with an arrow $\beta : R(A) \longrightarrow A$, which in this case is the same as a parengbra. Write $\mathbf{Set}^R$ for the category of R -algebras. We have the trivial identification

$$\mathcal{K} = \mathbf{Set}^R. \quad (1.7)$$

The free monad on an endofunctor is said to exist *pointwise* when the underlying functor from the category of algebras for the endofunctor has a left adjoint. It is an observation of Michael Barr (see [K3]) is that the free monad is then generated by that adjunction. Therefore, in our case, it follows that K is the pointwise free monad on the endofunctor R . As such, K is expressible as a power series; that is, there is a sequence $k_0, k_1, k_2, \dots$ of sets and a natural bijection

$$K(X) \cong k_0 + k_1 \times X + k_2 \times X^2 + k_3 \times X^3 + \dots \quad (1.8)$$

We can take the set k_n to consist of meaningfully bracketed words of length n . More precisely, we can find the sets k_n using the philosophy of clubs (see [K1] and [K4]). This leads to the expectation that the functor K can be recaptured from the free parengbra $k = K(1)$ on the singleton set 1 augmented by some “grounding” morphism. Let $\gamma : k \longrightarrow \mathbb{N}$ be the unique parengbra morphism taking the element

of the generating set 1 to the natural number 1 . This gives a grading of k according to the number of occurrences of the generator in the bracketed word; that is,

$$k_n = \gamma^{-1}(n). \quad (1.9)$$

For any set X , the iterative construction of $K(X)$ given by (i) and (ii) can be interpreted as a bijection

$$K(X) \cong X + K(X)^2 + K(X)^3 + K(X)^4 + K(X)^5 + \dots \quad (1.10)$$

These are the components of a natural isomorphism

$$K \cong 1_{\mathbf{Set}} + R \circ K \quad (1.11)$$

between endofunctors of $\mathbf{Set}$. One might hope that there is a category of “virtual sets” which are, to ordinary sets, as integers are to natural numbers; see [J2] for virtual species and [Sch]. Moreover, we can apply a virtual sets argument [J2] to the bijection of the last paragraph to obtain a “formula” for the k_n as mere sets. Put $Y = K(X)$ so that the bijection becomes

$$Y \cong X + Y^2 + Y^3 + Y^4 + Y^5 + \dots \cong X + Y^2 \times (1 - Y)^{-1}. \quad (1.12)$$

Multiplying on the right by $1 - Y$ and rearranging, we obtain the quadratic isomorphism

$$2Y^2 - (1 + X) \times Y + X \cong 0. \quad (1.13)$$

Solving by radicals and choosing the meaningful minus sign, we obtain

$$Y \cong \frac{1}{4}(1 + X - (1 - 6X + X^2)^{\frac{1}{2}}). \quad (1.14)$$

Applying the binomial theorem, we deduce a bijection

$$K(X) \cong X + X^2 + 3X^3 + 11X^4 + 45X^5 + \dots \quad (1.15)$$

If the reader is suspicious of virtual sets, this argument is at least valid in terms of generating functions and so gives the cardinalities of $k_0, k_1, k_2, k_3, k_4, k_5, \dots$ as $0, 1, 1, 3, 11, 45, \dots$. As an example, we see that for k_4 there are the eleven bracketings occurring as in the Tamari-Stasheff pentagon K_4 :

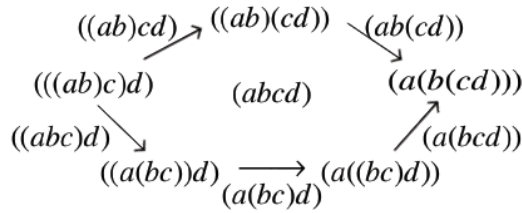


Figure 1

From this cell complex representation, we see that it is natural to define the *dimension* of an element $a \in K(X)$ to be $n - 2$ when n is the largest natural number for which β_n occurs in the iterative construction of a from elements of X . It is also natural to define the boundary $\partial(a)$ of $a \in K(X)$ to be the subset of $K(X)$ consisting of those elements obtained from a by meaningfully inserting precisely one further pair of brackets.

The Stasheff pentagon arose in homotopy theory, specifically, in studying the structure borne by loop spaces. This study was further advanced by May [M] who introduced the concept of operad. We shall see that our sequence $k = (k_n)$ of sets is indeed a planar (or non-permutative) operad.

2. Operads

Given two sequences p and q of sets, we define their *substitution product* $p \circ q$ to be the sequence of sets

$$(p \circ q)_n = \sum_{m_1 + \dots + m_r = n} p_r \times q_{m_1} \times \dots \times q_{m_r}. \quad (2.1)$$

We regard a sequence p of sets as a functor from the discrete category $\mathbb{N}$, whose objects are natural numbers, to the category **Set**. The substitution product defines a monoidal structure on the category **Set** ^{$\mathbb{N}$} of sequences of sets; the unit for substitution is the sequence u with u_1 a singleton set and $u_n = \emptyset$ for $n \neq 1$.

A (*planar*) operad t is a monoid for the substitution product on **Set** ^{$\mathbb{N}$} . Notice that the unit for such a monoid is an element $1 \in t_1$ and the multiplication amounts to a collection of functions

$$t_r \times t_{m_1} \times \dots \times t_{m_r} \longrightarrow t_{m_1 + \dots + m_r}, \quad (2.2)$$

whose value at $(\tau, \tau_1, \dots, \tau_r)$ is denoted by $\tau[\tau_1, \dots, \tau_r]$, called the result of *substituting* $\tau_1, \dots, \tau_r$ in τ , such that

$$1[\tau_1] = \tau_1 \quad (2.3)$$

$$\tau[1, \dots, 1] = \tau \quad (2.4)$$

$$\tau[\tau_1[\tau_{11}, \dots, \tau_{1s_1}], \dots, \tau_r[\tau_{r1}, \dots, \tau_{rs_r}]] = \tau[\tau_1, \dots, \tau_r][\tau_{11}, \dots, \tau_{1s_1}, \dots, \tau_{r1}, \dots, \tau_{rs_r}]. \quad (2.5)$$

An operad morphism is just a monoid morphism in **Set** ^{$\mathbb{N}$} with substitution tensor product.

Example 1 Let $\mathcal{A}$ be a monoidal category. Each object A of $\mathcal{A}$ gives rise to an operad $t = \mathcal{A}(A)$, where $t_n = \mathcal{A}(A^{\otimes n}, A)$ and substitution

$$\mathcal{A}(A^{\otimes r}, A) \times \mathcal{A}(A^{\otimes m_1}, A) \times \dots \times \mathcal{A}(A^{\otimes m_r}, A) \longrightarrow \mathcal{A}(A^{\otimes n}, A) \quad (2.6)$$

is given by $(f, g_1, \dots, g_r) \mapsto f \circ (g_1 \otimes \dots \otimes g_r)$. The distinguished element of t_1 is of course the identity morphism of A .

Example 2 Every operad t determines a strict monoidal category $\mathcal{V}_t$ as follows. The objects are the natural numbers and the homsets are given by

$$\mathcal{V}_t(m, n) = \sum_{m_1 + \dots + m_n = m} t_{m_1} \times \dots \times t_{m_n}. \quad (2.7)$$

Composition

$$\mathcal{V}_t(n, s) \times \mathcal{V}_t(m, n) \longrightarrow \mathcal{V}_t(m, s) \quad (2.8)$$

takes $(\tau_{n_1}, \dots, \tau_{n_s}, \tau_{m_1}, \dots, \tau_{m_n}) \in t_{n_1} \times \dots \times t_{n_s} \times t_{m_1} \times \dots \times t_{m_n}$ to

$$(\tau_{n_1}[\tau_{m_1}, \dots, \tau_{m_{n_1}}], \tau_{n_2}[\tau_{m_{n_1+1}}, \dots, \tau_{m_{n_1+n_2}}], \dots, \tau_{n_s}[\tau_{m_{n_1+\dots+n_{s-1}+1}}, \dots, \tau_{m_n}]) \quad (2.9)$$

where $m = m_1 + \dots + m_n$ and $n = n_1 + \dots + n_s$. The tensor product of $\mathcal{V}_t$ is given on objects by addition

and on homsets

$$\mathcal{H}_t(m, n) \times \mathcal{H}_t(r, s) \longrightarrow \mathcal{H}_t(m + r, n + s) \quad (2.10)$$

by $((\tau_{m_1}, \dots, \tau_{m_n}), \dots, (\tau_{r_1}, \dots, \tau_{r_s})) \mapsto (\tau_{m_1}, \dots, \tau_{m_n}, \dots, \tau_{r_1}, \dots, \tau_{r_s})$. Clearly we recapture the original operad t from the object $1 \in \mathcal{H}_t$ by the construction of Example 1. Furthermore, for any strict monoidal category $\mathcal{A}$ and $A \in \mathcal{A}$, the operad morphisms $\phi : t \longrightarrow \mathcal{A}(A)$ are in bijection with the strict monoidal functors $M : \mathcal{H}_t \longrightarrow \mathcal{A}$ with $M(1) = A$; the bijection is defined by putting $M(n) = A^{\otimes n}$ and taking the effect of M on homsets $\mathcal{H}_t(n, 1) \longrightarrow \mathcal{A}(M(n), M(1))$ to be ϕ_n .

Each operad gives rise to a monad (see [M] for the symmetric case). To see this, we identify each natural number $n \in \mathbb{N}$ with the set $\{0, 1, \dots, n-1\}$ and so obtain the natural sequence $nat : \mathbb{N} \longrightarrow \mathbf{Set}$ of sets. Left Kan extension along the functor nat gives a functor

$$Ser : \mathbf{Set}^{\mathbb{N}} \longrightarrow \mathbf{Set}^{\mathbf{Set}} \quad (2.11)$$

defined by $Ser(p) = P$, where

$$P(X) = \sum_{n \geq 0} p_n \times X^n \quad (2.12)$$

is the power series with coefficient sets p_n , $n \geq 0$. There is a (strict) monoidal structure on the category $\mathbf{Set}^{\mathbf{Set}}$ of endofunctors of $\mathbf{Set}$ defined by composition of functors. A standard calculation with power series shows that Ser is a strong monoidal functor; that is, it coherently preserves the monoidal structures:

- if $P = Ser(p)$ and $Q = Ser(q)$ then $P \circ Q \cong Ser(p \circ q)$; and,
- $Ser(u)$ is isomorphic to the identity functor.

It follows that monoids t in $\mathbf{Set}^{\mathbb{N}}$ are taken to monads $T = Ser(t)$ on the category $\mathbf{Set}$. An *algebra for the operad t* is defined to be an Eilenberg-Moore algebra for the monad $T = Ser(t)$.

Besides substitution product, there are of course other useful operations on sequences of sets. The *sum* $p + q$ of sequences p and q is given pointwise by

$$(p + q)_n = p_n + q_n \quad (2.13)$$

(disjoint union of sets); this is the coproduct of p and q in the category $\mathbf{Set}^{\mathbb{N}}$ and, likewise, infinite sums can be considered. The *convolution product* $p * q$ of p and q is given by

$$(p * q)_n = \sum_{r+s=n} p_r \times q_s. \quad (2.14)$$

It is easy to see that the functor $Ser : \mathbf{Set}^{\mathbb{N}} \longrightarrow \mathbf{Set}^{\mathbf{Set}}$ takes sum to coproduct and convolution product to product.

We can identify each set z with the sequence z given by $z_0 = z$ and $z_n = 0$ for all $n > 0$. Then each sequence p of sets can be decomposed as a power series

$$p \cong \sum_{n \geq 0} p_n * u^{*n} \quad (2.15)$$

where u^{*n} is the n -fold convolution power of the unit u for substitution product.

Example 3 Let u^* denote the terminal sequence of sets; that is, each set in the sequence is a singleton. This clearly has a unique operad structure. The monad $Ser(u^*)$ on **Set** is given by the full geometric series

$$X^* = \sum_{n \geq 0} X^n . \quad (2.16)$$

The elements of X^* are written as *words* in the *alphabet* X . The algebras for the terminal operad are the Eilenberg-Moore algebras for the geometric series monad, and so are monoids (in **Set** with cartesian product as monoidal structure).

Example 4 Let z be any set. Consider the sequence $z+u$ of sets given by

$$(z+u)_n = \begin{cases} z & \text{for } n=0 \\ 1 & \text{for } n=1 \\ 0 & \text{for } n>1 \end{cases} . \quad (2.17)$$

There is a unique operad structure on $z+u$ for which the substitution $(z+u)_1 \times (z+u)_0 \longrightarrow (z+u)_0$ is the second projection. The monad $Ser(z+u)$ takes X to the monic polynomial $z+X$ of degree 1. The monad structure on $Ser(z+u)$ is induced by the canonical monoid structure on the arbitrary set z with respect to the coproduct as tensor product on **Set**. The algebras for the operad $z+u$ are sets A together with a function $z+A \longrightarrow A$ whose restriction to A is the identity; that is, the algebras amount to functions $z \longrightarrow A$ out of z . The category of algebras is the category $z/\mathbf{Set}$ of sets under z .

Example 5 Any monad T on **Set** admits a distributive law [Bk] with the monad $Ser(z+u)$ of Example 4; that is, there is a natural transformation $\lambda : Ser(z+u) \circ T \longrightarrow T \circ Ser(z+u)$ satisfying axioms ensuring that T lifts to a monad on $Ser(z+u)$ -algebras, and that $T \circ Ser(z+u)$ gains a monad structure whose algebras are the same as the algebras for the lifted monad; in this case, the category of these algebras is just the category $z/\mathbf{Set}^T$ of T -algebras under z . The component $\lambda_X : z+T(X) \longrightarrow T(z+X)$ of λ at X is constructed from the unit of T at z and T of the inclusions of z and X in $z+X$. Suppose now that $T = Ser(t)$ for some operad t . Then $T \circ Ser(z+u) = Ser(t \circ (z+u))$ for the operad $t \circ (z+u)$ calculated as follows:

$$\begin{aligned} (T \circ Ser(u+z))(X) &= \sum_{n \geq 0} t_n \times (z+X)^n = \sum_{n \geq 0} t_n \times \sum_{0 \leq m \leq n} \binom{n}{m} z^{n-m} \times X^m \\ &= \sum_{\substack{m \geq 0 \\ r \geq 0}} \binom{r+m}{m} t_{r+m} \times z^r \times X^m = Ser(t \circ (z+u))(X) \end{aligned} \quad (2.18)$$

where $(t \circ (z+u))_m = \sum_{0 \leq r} \binom{r+m}{m} t_{r+m} \times z^r$. We shall not bother to explicitly describe the substitution

operation of $t \circ (z + u)$ except in the special case of the next example.

Example 6 As a particular case of Example 5, take $t = u^*$ to be the terminal sequence as discussed in Example 3. Then $T \circ \text{Ser}(z + u)$ is the monad given by

$$(T \circ \text{Ser}(z + u))(X) = (z + X)^* . \quad (2.19)$$

The operad $t \circ (z + u) = (z + u)^*$ is given by taking $(z + u)_m^*$ to consist of those elements of the free monoid $(z + 1)^*$ which are words in elements of z and the symbol 0 with exactly m occurrences of 0.

For we have the natural bijection

$$(z + X)^* \cong \sum_{m \geq 0} (z + u)_m^* \times X^m \quad (2.20)$$

$$a_0 x_1 a_1 x_2 a_2 \dots a_{m-1} x_m a_m \leftrightarrow ((a_0 0 a_1 0 a_2 \dots a_{m-1} 0 a_m), (x_1, x_2, \dots, x_m))$$

where the $a_0, \dots, a_m \in z^*$ and $x_1, \dots, x_m \in X$. We have the substitution functions

$$(z + u)_r^* \times (z + u)_{m_1}^* \times \dots \times (z + u)_{m_r}^* \longrightarrow (z + u)_{m_1 + \dots + m_r}^* \quad (2.21)$$

$$(a, b_1, \dots, b_r) \mapsto a_0 b_1 a_1 \dots a_{r-1} b_r a_r$$

where $a = a_0 0 a_1 0 \dots a_{r-1} 0 a_r$. The distinguished element of $(z + u)_1^*$ is of course $0 \in 1$.

3. Well-formed words

Proposition 1 below will follow from Proposition 2 which will follow, in turn, from a discussion of well-formed words. The criterion for when an expression in a universal algebra, written in Polish or Lukasiewicz notation, is well formed I learned from Samuel Eilenberg who caused it, I believe, to be an exercise in Bourbaki. Now I see that this classical result was the starting point for Tamari's paper [Ta], which went on to provide a criterion (in terms of cyclic words) useful for enumeration.

Proposition 1 *Free monads on power series endofunctors of **Set** are all of the form $\text{Ser}(t)$ for an operad t .*

Let P be a power series endofunctor of **Set** with coefficients given by the sequence p of sets. We regard the elements of p_n as n -ary operations. A P -algebra is a set A together with, for each $n \in \mathbb{N}$, and each $\omega \in p_n$, a function $A^n \longrightarrow A$ which is also denoted by ω . The left adjoint of underlying functor $\mathbf{Set}^P \longrightarrow \mathbf{Set}$ can be described iteratively as we did earlier for the case $P = R$. Then the adjunction generates the pointwise free monad on P . We shall give another construction related to Example 6.

Consider the free monoid $(P(1) + X)^*$ on the set $P(1) + X$ and regard the set $\mathbb{Z}$ of integers as a monoid under addition. Let $(\bar{\cdot}) : (P(1) + X)^* \longrightarrow \mathbb{Z}$ be the monoid morphism given on generators by:

$$\bar{x} = -1 \quad \text{for all } x \in X \quad \text{and} \quad (3.1)$$

$$\bar{\omega} = n - 1 \quad \text{for all } \omega \in p_n . \quad (3.2)$$

Call $a \in (P(1) + X)^*$ *well formed* when

- i. $\bar{a} = -1$ and

ii. $a = bc$ implies $\bar{c} < 0$.

Let $W_p(X)$ denote the subset of $(P(1) + X)^*$ consisting of the well-formed words. We can equip $W_p(X)$ with the structure of P -algebra by defining

$$\omega(a_1, \dots, a_n) = \omega a_1 \dots a_n \quad (3.3)$$

for all $\omega \in p_n$ and $a_1, \dots, a_n \in W_p(X)$. To see that $\omega a_1 \dots a_n$ is well formed notice that

- $\overline{\omega a_1 \dots a_n} = \bar{\omega} + \bar{a}_1 + \dots + \bar{a}_n = (n-1) + (-1) + \dots + (-1) = -1$ and
- if $a_i = bc$ then $\overline{ca_{i+1} \dots a_n} = \bar{c} + \bar{a}_{i+1} + \dots + \bar{a}_n = \bar{c} + (-1) + \dots + (-1) = \bar{c} + (i-n) < 0$.

Proposition 2 $W_p(X)$ is the free P -algebra on the set X .

Proof Given a function $f: X \longrightarrow A$ into a P -algebra A , we must show that there exists a unique extension of f to a P -algebra morphism $g: W_p(X) \longrightarrow A$. That is, we must show that the equations

$$g(x) = f(x) \quad \text{for } x \in X \quad (3.4)$$

$$g(\omega a_1 \dots a_n) = \omega(g(a_1), \dots, g(a_n)) \quad \text{for } \omega \in p_n \text{ and } a_1, \dots, a_n \in W_p(X) \quad (3.5)$$

uniquely determine g satisfying the equations. To define $g(a)$ for $a \in W_p(X)$, we use induction on the number r of occurrences of elements of $P(1)$ in the word a . By well formedness, if $r = 0$ then $a \in X$ and the definition $g(a) = f(a)$ is forced. Suppose $r > 0$. By well formedness, a must have length > 1 ; so put $a = tb$ where $t \in X + P(1)$. But $\bar{b} < 0$ and $\bar{t} + \bar{b} = -1$; so $\bar{t} > -1$, whence $t \in p_n$ for some $n = -\bar{b}$ and we can write $b = c_1 \dots c_n$ where $\bar{c}_1 = \dots = \bar{c}_n = -1$. Since a is well formed, it follows that $c_1, \dots, c_n \in W_p(X)$. Define $g(a) = \omega(g(c_1), \dots, g(c_n))$ as we are forced to, and can do since each c_i has fewer than r occurrences of elements of $P(1)$. It remains to prove that g is a P -algebra morphism but that is a direct inductive argument. **QED**

To construct the operad from W_p , notice that $\mathbb{N}$ becomes a P -algebra by taking each n -ary operation to be n -fold addition. Then we obtain a P -algebra morphism

$$\gamma: W_p(1) \longrightarrow \mathbb{N} \quad (3.6)$$

which restricts to the function $1 \longrightarrow \mathbb{N}$ whose value at the one element 0 of 1 is $1 \in \mathbb{N}$. So $\gamma(a)$ is the number of occurrences of the element 0 of 1 in the word a . Let $t_n = \gamma^{-1}(n)$ be the fibre of γ over n . If we put $z = P(1) = p_0 + p_1 + p_2 + \dots$ we see that W_p is a submonad of the monad $Ser(z+u)^*$ described in Example 6. Furthermore, the gradings are respected so that t is a suboperad of $(z+u)^*$.

Corollary 3 The operad t is free on the sequence p of sets. The monad $T = Ser(t)$ is free on the endofunctor P .

Proof Suppose o is an operad and $\theta_n: p_n \longrightarrow o_n$ are functions for all $n \geq 0$. We can make $\sum_{m \geq 0} o_m$ into a P -algebra by defining, for each $\omega \in p_n$, a function

$$\omega : (\sum_{m \geq 0} o_m)^n = \sum_{m_1, \dots, m_n} o_{m_1} \times \dots \times o_{m_n} \longrightarrow \sum_{m \geq 0} o_m \quad (3.7)$$

by $\omega(v_1, \dots, v_n) = \theta_n(\omega)[v_1, \dots, v_n]$. By Proposition 2, there is a unique P -algebra morphism $\phi : W_p(1) \longrightarrow \sum_{m \geq 0} o_m$ determined by $\phi(0) = 1 \in u_1$. This clearly respects the gradings and so gives functions $\phi_n : t_n \longrightarrow o_n$. Since ϕ is a P -algebra morphism, the functions ϕ_n for $n \in \mathbb{N}$ commute with the substitution operations and ϕ_1 preserves the distinguished object. So (ϕ_n) is the unique operad morphism extending (θ_n) .

The free monad on the endofunctor P on **Set** is the monad generated by the underlying functor $\mathbf{Set}^P \longrightarrow \mathbf{Set}$ and its left adjoint. By Proposition 2, this left adjoint is provided by W_p . It is easily seen that the monad structure transports across our isomorphism $W_p \cong \text{Ser}(t)$ to the monad structure induced by the operad t . **QED**

In particular, taking the sequence r of sets given by $r_n = 0$ for $n = 0, 1$ and $r_n = 1$ for $n \geq 2$, so that the power series endofunctor $\text{Ser}(r)$ is R , we obtain a parengebra isomorphism

$$W_r(X) \cong F(X) \quad (3.8)$$

an operad structure on k producing the monad structure on $K = \text{Ser}(k)$. A topological version of k is the Stasheff A_∞ -operad.

A monad T on a category $\mathcal{C}$ is *cartesian* (see [Bén]) when the endofunctor $T : \mathcal{C} \longrightarrow \mathcal{C}$ preserves pullbacks and the multiplication $\mu : T \circ T \longrightarrow T$ and unit $\eta : 1_{\mathcal{C}} \longrightarrow T$ are cartesian natural transformations. A natural transformation $\theta : H \longrightarrow K : \mathcal{C} \longrightarrow \mathcal{D}$ is *cartesian* when, for all morphisms $f : U \longrightarrow V$ in $\mathcal{C}$, the commutative square

$$\begin{array}{ccc} HU & \xrightarrow{\theta_U} & KU \\ Hf \downarrow & & \downarrow Kf \\ HV & \xrightarrow{\theta_V} & KV \end{array} \quad (3.9)$$

is a pullback in $\mathcal{D}$.

Corollary 4 *For the free operad t on a sequence p of sets, the monad $T = \text{Ser}(t)$ is cartesian.*

Proof It is well known that the free monoid monad $(\)^*$ on **Set** (see Example 3) is cartesian (it is even true also for **Set** replaced by a topos with natural numbers object; see [Bén]). It is easy to see that the monad $\text{Ser}(z+u)$ of Example 4 is cartesian. It follows from this that the monad $X \mapsto (z+X)^*$ of Example 6 is cartesian. In particular, this is true with $z = P(1)$. The pullbacks proving cartesianness of the monad generated by the adjunction of Proposition 2 (that is, W_p and its forgetful right adjoint) are a consequence of those for $X \mapsto (P(1)+X)^*$ and their closure under wellformedness. **QED**

For recent developments see [GK].

4. Free monoidal categories

We shall make a connection with free monoidal categories on certain³ tensor schemes [JS]. A *tensor scheme* consists of a set whose elements are called *objects* and a graph whose vertices are words of objects and whose edges are called *morphisms*. Words $D_1 \dots D_n$ of objects are written $D_1 \otimes \dots \otimes D_n$. Here we restrict to the case where the target of each morphism is a word of length 1.

Take $p \in \mathbf{Set}^{\mathbb{N}}$. Consider the tensor scheme $\mathcal{D}_p$ with one object D and one morphism

$$\omega : D^{\otimes n} \longrightarrow D \quad (4.1)$$

for each natural number n and each $\omega \in p_n$. We represent the morphism ω by a planar string diagram where there are n input strings above the node ω and 1 output string below⁴.

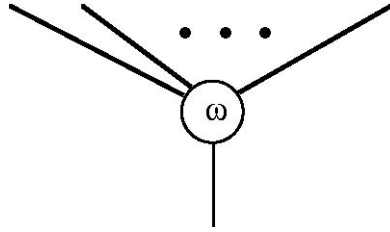


Figure 2

Let $\mathcal{D}_p^*$ denote the free strict monoidal category on the tensor scheme $\mathcal{D}_p$: the objects are of the form $D^{\otimes n}$ for $n \in \mathbb{N}$ and the morphisms can be identified with the deformation classes of planar string diagrams generated by those representing morphisms of $\mathcal{D}_p$. Write $\mathbf{Mon}(\mathcal{D}_p^*, \mathbf{Set})$ for the category of (strong) monoidal functors from the monoidal category $\mathcal{D}_p^*$ to $\mathbf{Set}$ with its cartesian monoidal structure. Clearly there is an equivalence of categories

$$\mathbf{Set}^p \simeq \mathbf{Mon}(\mathcal{D}_p^*, \mathbf{Set}). \quad (4.2)$$

Corollary 5 *If t is the free operad on the sequence p of sets then there is a monoidal isomorphism*

$$\mathcal{D}_p^* \cong \mathcal{V}_t \quad (4.3)$$

taking D to 1 and $\omega : D^{\otimes n} \longrightarrow D$ to $\omega 0 \dots 0 \in t_n = \mathcal{V}_t(n, 1)$ for $\omega \in p_n$.

Proof We must prove that, given a strict monoidal category $\mathcal{A}$, an object A of $\mathcal{A}$, and a morphism $f_\omega : A^{\otimes n} \longrightarrow A$ for each $\omega \in p_n$, there exists a unique strict monoidal functor $M : \mathcal{V}_t \longrightarrow \mathcal{A}$ with $M(1) = A$ and $M(\omega 0 \dots 0) = f_\omega$. Since t is the free operad on p , there is a unique operad morphism $\phi : t \longrightarrow \mathcal{A}(A)$ (see Example 1) whose restriction to p is given by $\omega \mapsto f_\omega$. By Example 2, this operad morphism determines a strict monoidal functor M as required. **QED**

³ One might call them opetopic [BD] tensor schemes or multigraphs.

⁴ Here we take the progressive direction to be downward which is opposite to [JS] but the same as [St2].

The arrows of $\mathcal{D}_p^*$ can be identified with p -labelled (planar) forests. To see this, recall from [JS] the general string diagrams are built from the generators in $\mathcal{D}_p$. We take some arrows from $\mathcal{D}_p$



Next we stack such diagrams vertically splicing each lower loose string of the upper diagram with precisely one upper loose string of the lower diagram. The diagram of Figure 3 is composable with the diagram of Figure 4 and the composite is represented in Figure 5.



But such a planar diagram can be replaced by a more combinatorial structure. We can represent the composite in Figure 5 by the diagram of Figure 6 where we have removed the bottom loose strings, put square nodes on the top loose strings, and shortened the strings so that they connect nodes of consecutive height.

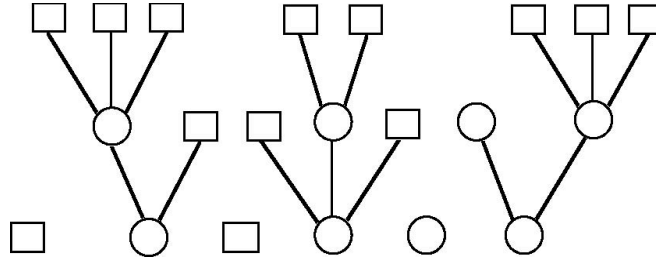


Figure 6

We must remember that, although we have omitted it from the diagram, each round node with n strings attached at top is labelled by an element of p_n . The structure that arises in this way is precisely a *planar forest*; that is, a functor $f:[k]^{op} \longrightarrow \Delta$ where Δ is the category whose objects are the linearly ordered sets $[k] = \{0, 1, \dots, k\}$ for $k \geq -1$ and whose morphisms are order-preserving functions. Such a functor gives linearly ordered sets $f(i)$ for $0 \leq i \leq k$, whose elements are called *vertices* of f of *height* i , and order-preserving functions $f_i: f(i+1) \longrightarrow f(i)$ for $0 \leq i < k$. In our example of Figure 6,

$$\begin{aligned} f(0) &= [5], f(1) = [6], f(2) = [7] \\ f_0(0) &= f_0(1) = 1, f_0(2) = f_0(3) = f_0(4) = 3, f_0(5) = f_0(6) = 5 \\ f_1(0) &= f_1(1) = f_1(2) = 0, f_1(3) = f_1(4) = 3, f_1(5) = f_1(6) = f_1(7) = 6. \end{aligned}$$

A forest $f:[k]^{op} \longrightarrow \Delta$ is called a *tree* when $f(0) = [0]$. Each forest can be identified with a linearly ordered set of component trees: the number of component trees is n where $f(0) = [n-1]$.

For a forest $f:[k]^{op} \longrightarrow \Delta$ of *height* k , a vertex $v \in f(i)$ is called a *leaf* when the fibre of f_i over v is empty. Notice in our Figure 6 example that the square nodes are all leaves; but there are also two round nodes that are leaves. Let p be a sequence of sets. A *labelling* of a forest f in p assigns to each vertex $v \in f(i)$, which is not a leaf, an element $\omega(v) \in p_n$ where $n > 0$ is the cardinality of the fibre of f_i over v , and assigns to some leaves $v \in f(i)$, which are not of height k , an element $\omega(v) \in p_0$. A leaf which is not labelled will be called *fallen*; so all leaves of height k are fallen. A forest together with a labelling in p will be called a *p-forest*.

This leads to our next view of the strict monoidal category $\mathcal{D}_p^*$. We see that the morphisms $D^{\otimes m} \longrightarrow D^{\otimes n}$ can be identified with p -forests having m fallen leaves and n vertices of height 0. The tensor product of two p -forests is given by placing them next to each other in the plane; we shall not describe it combinatorially. Composition is given by *grafting*: we shall just give an example. The p -forest in Figure 7 is composable with that in Figure 8. Their composite is the p -forest shown in Figure 9 in which the component trees of the upper forest are grafted on at the fallen leaves of the lower forest. From this, we deduce another view of the free operad t on p in which elements of t_n are identified with p -trees with n fallen leaves.

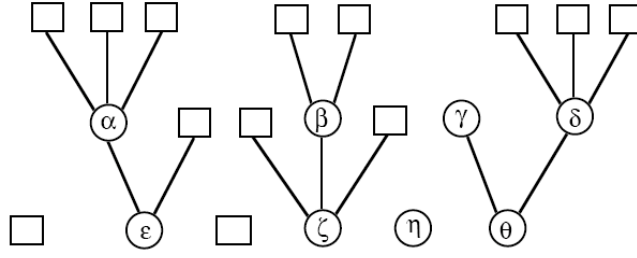


Figure 7

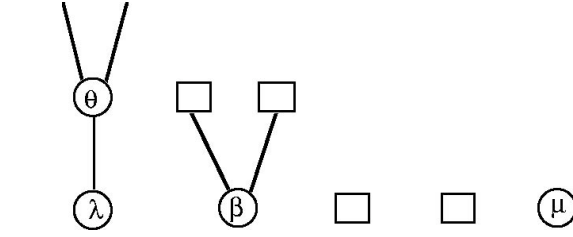


Figure 8

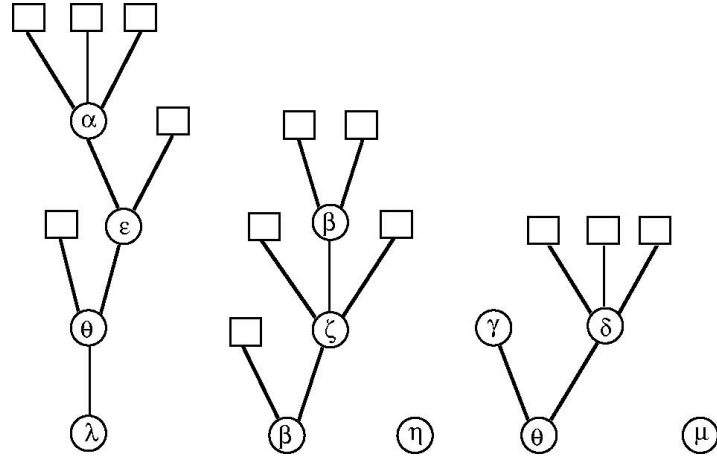


Figure 9

A description of $\mathcal{D}_p^*$ in the spirit of computer science can be obtained using *rewrite systems* (see [St2] for example).

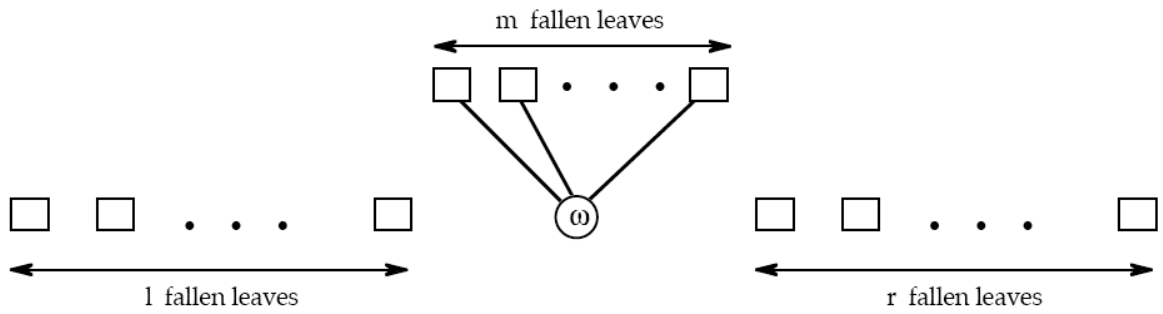


Figure 10

The basic p -forest shown in Figure 10 can be identified with triplets (l, ω, r) where l, m, r are natural numbers and $\omega \in p_m$. Consider the directed graph $\mathcal{G}_p$ whose vertices are natural numbers

and whose edges are the expressions

$$(\omega, r) : a \longrightarrow b \quad (4.4)$$

where a, b, r are natural numbers such that $r < b \leq a + 1$ and $\omega \in p_m$ where $m = a - b + 1$. The idea is that each $\omega \in p_m$ is regarded as the name of a (first-order) rewrite rule and $(\omega, r) : a \longrightarrow b$ represents an allowable application of the rule; the numbers $l = b - 1 - r$ and r are the left and right “whiskers”, respectively. We introduce some (second-order) rewrite rules on directed paths in the graph $\mathcal{G}_p$.

$$\frac{a \xrightarrow{(\omega, r)} b \xrightarrow{(\omega', r')} c}{a \xrightarrow{(\omega', r' + a - b)} a - b + c \xrightarrow{(\omega, r)} c} \quad \text{for } r < r' \quad (4.5)$$

It is easy to see that the p -forests corresponding to the top and bottom of the rewrite rule are equal. Two directed paths in $\mathcal{G}_p$ with same source and target are said to be *equivalent* when there is a sequence of applications of the (second-order) rewrite rules which takes one path to the other. A directed path

$$a_0 \xrightarrow{(\omega_1, r_1)} a_1 \xrightarrow{(\omega_2, r_2)} \dots \xrightarrow{(\omega_n, r_n)} a_n \quad (4.6)$$

is said to be in *normal form* when $r_i \geq r_{i+1}$ for $1 \leq i \leq n$. Each directed path is equivalent to a unique path in normal form; indeed, the normal form can be achieved by directed applications of the (second-order) rewrite rules (the proof of confluence and termination is similar to the case considered in [St2]).

In this way, we obtain a strict monoidal category isomorphic to $\mathcal{D}_p^*$. The objects are natural numbers and the arrows are equivalence classes of directed paths in $\mathcal{G}_p$. The composition is induced on equivalence classes by concatenation of paths. The tensor product $a \otimes b$ is given on objects a and b by addition $a + b$ of natural numbers. The functor $c \otimes -$ is given on morphisms by

$$c \otimes [(\omega_n, r_n) \circ \dots \circ (\omega_1, r_1)] = [(\omega_n, r_n) \circ \dots \circ (\omega_1, r_1)] \quad (4.7)$$

and the functor $- \otimes d$ is given on morphisms by

$$[(\omega_n, r_n) \circ \dots \circ (\omega_1, r_1)] \otimes d = [(\omega_n, r_n + d) \circ \dots \circ (\omega_1, r_1 + d)]; \quad (4.8)$$

the tensor product $\pi \otimes \rho : a \otimes c \longrightarrow b \otimes d$ of two arrows $\pi : a \longrightarrow b$ and $\rho : c \longrightarrow d$ is then given by either route around the following square, the commutativity of which is precisely what we have achieved by our equivalence relation on paths.

$$\begin{array}{ccc} a \otimes c & \xrightarrow{\pi \otimes c} & b \otimes c \\ a \otimes \rho \downarrow & & \downarrow b \otimes \rho \\ a \otimes d & \xrightarrow{\pi \otimes d} & b \otimes d \end{array} \quad (4.9)$$

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